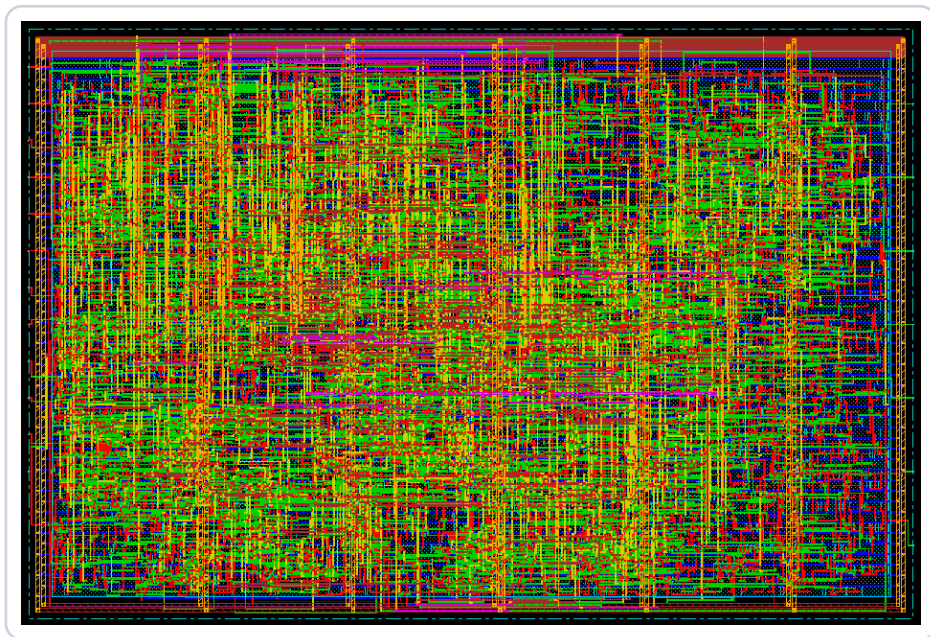


UART-Based Master-Slave System Controller

A multi-clock-domain digital system, taken from RTL to GDSII

TSMC 0.13 μm | SpyGlass | Design Compiler | SoC Encounter | Formality



99.53%

scan test coverage

0

unsynchronized CDC crossings

3 of 3

Formality checks passed

3.431 ns

post-route setup WNS

0

DRC / antenna / connectivity violations

240 × 160 μm

die size

Mohamed Aboualy

Electronics and Electrical Communications Engineering, Faculty of Engineering, Cairo University

github.com/mo-aboualy/UART-Based-Master-Slave-System-Controller | October 2026

Contents

1 Introduction	3
1.1 Project overview	3
1.2 Specification summary	3
1.3 Results at a glance	3
2 System architecture	4
2.1 Block diagram	4
2.2 Clocks and resets	5
2.3 Signal names in the block diagram	5
3 Protocol, registers and ALU	5
3.1 Command protocol	5
3.2 Register file map	6
3.3 ALU function encoding	6
4 RTL design	6
4.1 UART transmitter	6
4.2 UART receiver	6
4.3 System controller	7
4.4 Register file and ALU	7
4.5 Clock management	7
4.6 Clock domain crossing strategy	7
5 Functional verification	8
5.1 Console results	9
5.2 Waveforms	9
6 Lint and CDC / RDC sign-off	11
7 Logic synthesis	11
7.1 Area and power	11
7.2 Timing	12
8 Design for test	12
9 Place and route	12
9.1 Flow steps	13
9.2 Post-route results	14
9.3 Clock tree	15
10 Formal equivalence checking	15
11 Cross-stage summary and limitations	15
11.1 Summary across the flow	15
11.2 Limitations and observations	16
11.3 Possible extensions	16
12 Conclusion	16
Appendix A Repository structure	17
Appendix B Tools and technology	17

1 Introduction

1.1 Project overview

This report documents the design, verification and physical implementation of a **UART-based master-slave system controller**: a digital system that receives byte-oriented commands over a serial UART link, executes them on a 16-register file or a signed arithmetic and logic unit (ALU), and returns the results over the same link. The design spans two asynchronous clock domains and is taken through the complete RTL-to-GDSII flow on a TSMC 0.13 μm standard-cell library.

Every netlist transformation is checked against the golden RTL with formal equivalence, and the clock domain crossings are signed off with structural CDC and reset-domain-crossing analysis. The report follows the flow in order: architecture and RTL (sections 2 to 4), functional verification (5), lint and CDC sign-off (6), logic synthesis (7), design for test (8), place and route (9), formal equivalence (10), and a cross-stage summary with limitations (11).

1.2 Specification summary

Item	Value
Clock domains	REF_CLK 50 MHz (control, register file, ALU) and UART_CLK 3.6864 MHz (UART), mutually asynchronous
Serial interface	115200 baud default, 8 data bits, even parity enabled, 1 stop bit; parity and prescale are software-configurable
Command set	Register write (0xAA), register read (0xBB), ALU operation with operands (0xCC), ALU operation without operands (0xDD)
Register file	16 registers of 8 bits; four reserved for operand A, operand B, UART configuration and the TX division ratio
ALU	8-bit signed operands, 16-bit result, 16 functions in four groups: arithmetic, logic, comparison and shift
Clock management	Programmable clock dividers for the UART RX and TX clocks, latch-based clock gating for the ALU
Clock domain crossing	Reset synchronizers, enable-pulse data synchronizer, 8-entry asynchronous FIFO with Gray-coded pointers
Technology	TSMC 0.13 μm (scmetro_tsmc_cl013g, regular Vt), slow / typical / fast corners
Tools	ModelSim/Questa, SpyGlass, Design Compiler (O-2018.06-SP1), Formality, Cadence SoC Encounter 8.1

1.3 Results at a glance

Stage	Key result
Functional simulation	10 of 10 self-checking tests pass: configuration, register write and read, ALU with and without operands, frame integrity
Lint, CDC and RDC	0 unsynchronized scalar or vector crossings, 0 reset domain crossings, 0 glitches on synchronized control paths
Synthesis	25,680 μm^2 , 2,282 cells, 0.363 mW at the slow corner, all setup and hold paths met, no constraint violations
Design for test	4 scan chains, 374 scan cells, 99.53% estimated test coverage
Place and route	Setup WNS +3.431 ns, hold WNS +0.019 ns, 0 violating paths of 1,062; geometry, connectivity and antenna checks clean; 240.47 $\times$ 160.47 μm die
Formal equivalence	RTL vs. synthesis 369 / 369, RTL vs. DFT 378 / 378, RTL vs. routed netlist 378 / 378: all Verification SUCCEEDED

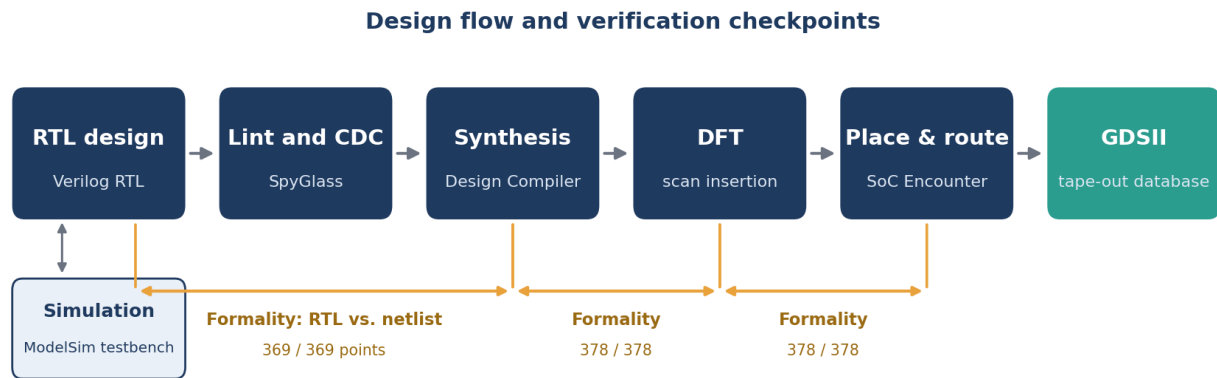


Figure 1. Design flow with simulation and the three formal equivalence checkpoints.

2 System architecture

2.1 Block diagram

A master (the testbench) sends command frames on `RX_IN`. The UART receiver deserializes each frame in the `UART_CLK` domain and a data synchronizer moves the byte into the `REF_CLK` domain. The system controller decodes the command and drives either the register file or the ALU. Results are written into an asynchronous FIFO, read out in the UART clock domain, and transmitted on `TX_OUT`.

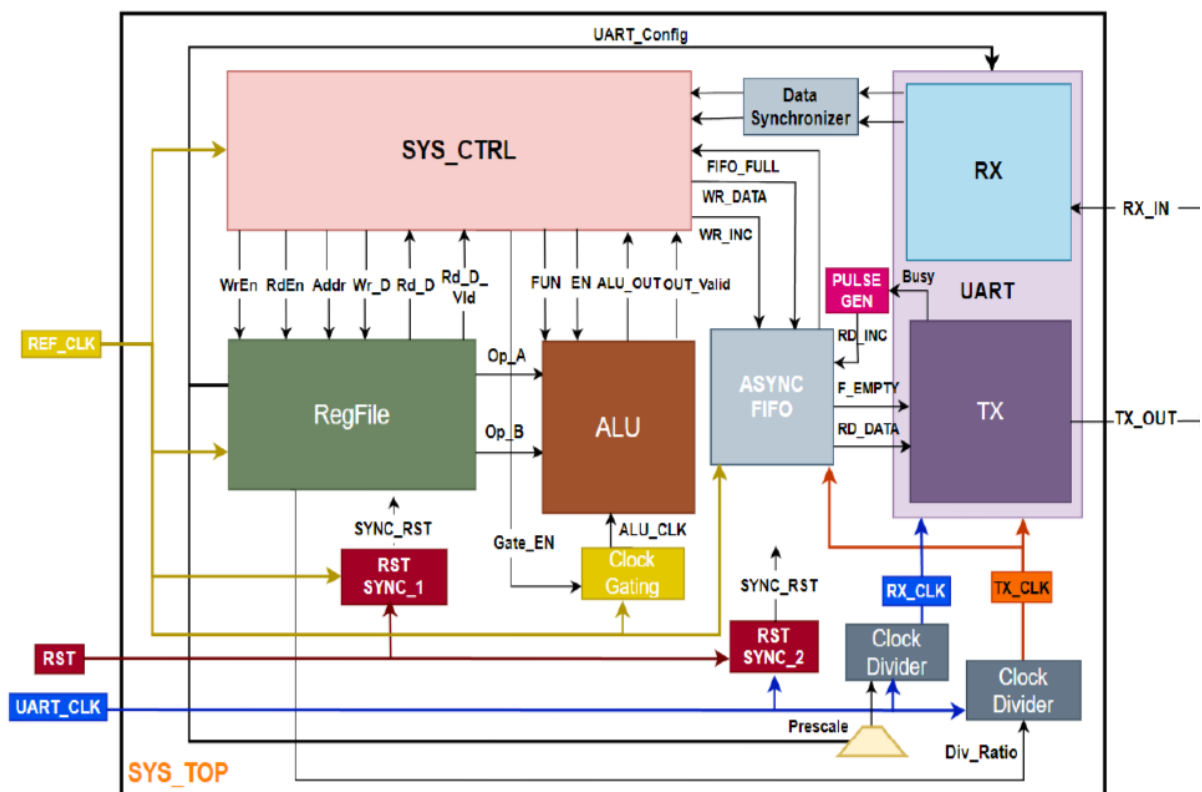


Figure 2. SYS_TOP block diagram. Clock paths are colored by domain; RX_CLK and TX_CLK come from the two clock dividers.

Group	Blocks
Clock domain 1 (REF_CLK)	Register file, ALU, clock gating, system controller
Clock domain 2 (UART_CLK)	UART TX, UART RX, pulse generator, RX and TX clock dividers
Synchronizers	Reset synchronizer (one per domain), data synchronizer, asynchronous FIFO

2.2 Clocks and resets

REF_CLK (50 MHz) clocks the system controller, register file and the write side of the FIFO. The ALU runs on ALU_CLK, a gated copy of REF_CLK that is enabled only while an operation is in flight. UART_CLK (3.6864 MHz) is the source for both UART clocks. The RX clock divider ratio is chosen from the prescale field of REG2 through a small lookup (32 maps to /1, 16 to /2, 8 to /4, 4 to /8), and the TX clock is UART_CLK divided by REG3. With the defaults, RX_CLK runs at 3.6864 MHz (32x oversampling of the bit rate) and TX_CLK at 115.2 kHz, which gives 115200 baud.

The global active-low reset is synchronized separately into each domain by a two-stage reset synchronizer: assertion is asynchronous, de-assertion is synchronous to the local clock.

2.3 Signal names in the block diagram

The labels in the figure use short names; the table maps them to the RTL signals.

Figure label	RTL signal	Figure label	RTL signal
UART_Config	REG2 (UART_CONFIG)	Gate_EN	CLK_en
Prescale	REG2[7:2]	FUN	ALU_FUNC
Div_Ratio	REG3 (TX_DIV_Ratio)	EN	decoded in the ALU from ALU_FUNC[3:2]
Wr_D / Rd_D	WR_Data / RD_Data	FIFO_FULL	W_full
Rd_D_Vld	RD_Valid	WR_INC / WR_DATA	W_inc / WR_DATA_fifo
SYNC_RST	RST_Domain_1 / RST_Domain_2	RD_INC / F_EMPTY	R_inc / R_empty
Busy	busy (UART TX)	RD_DATA	TX_P_data

3 Protocol, registers and ALU

3.1 Command protocol

Every frame is a single UART byte. The first frame of a command selects the operation. The typical sequence is to write the configuration registers (0x2, 0x3) and then issue register and ALU commands.

Command	Opcode	Frames	Layout (in order sent)	Response
Register file write	0xAA	3	0xAA, address, data	none
Register file read	0xBB	2	0xBB, address	1 byte (register value)
ALU operation with operands	0xCC	4	0xCC, operand A, operand B, ALU_FUNC	2 bytes (low byte, then high byte)
ALU operation without operands	0xDD	2	0xDD, ALU_FUNC	2 bytes (low byte, then high byte)

0xCC writes operands A and B into registers 0x0 and 0x1; 0xDD reuses whatever those registers already hold. A **UART frame** consists of a start bit (0), 8 data bits sent LSB first, an optional parity bit and a stop bit (1). The parity bit is $\text{data} \wedge \text{PAR_TYPE}$, so PAR_TYPE = 0 selects even parity.

3.2 Register file map

Address	Name	Function	Reset value
0x0	REG0	ALU operand A	0x00
0x1	REG1	ALU operand B	0x00
0x2	REG2	UART configuration: [0] parity enable, [1] parity type, [7:2] prescale	0x81 (parity on, even, prescale 32)
0x3	REG3	TX clock division ratio	0x20 (32)
0x4 and up		General-purpose storage	0x00

3.3 ALU function encoding

ALU_FUN[3:2] selects the functional unit (arithmetic, logic, comparison or shift) and ALU_FUN[1:0] selects the operation within it. Arithmetic is signed and the result is 16 bits wide; division by zero returns 0.

ALU_FUN	Operation	ALU_FUN	Operation
0000	A + B	1000	NOP (output 0)
0001	A - B	1001	A == B (1 if equal)
0010	A × B	1010	A > B (2 if true)
0011	A / B (0 when B = 0)	1011	A < B (3 if true)
0100	A AND B	1100	A >> 1
0101	A OR B	1101	A << 1
0110	A NAND B	1110	B >> 1
0111	A NOR B	1111	B << 1

4 RTL design

The RTL is written in Verilog, one module per file, organized by block. This section summarizes the behavior and the main design decisions of each block.

4.1 UART transmitter

An FSM with five states (Idle, Start, Data, Parity, Stop) controls a four-way output multiplexer that selects the start bit, the serialized data bit, the parity bit or the stop bit. The serializer shifts the byte out LSB first and raises `ser_done` after the eighth bit. The parity calculator computes $\text{data} \wedge \text{PAR_type}$ when a byte is accepted. In Idle the multiplexer selects the stop-bit value so the line rests high. While the FSM is busy, the `busy` flag is converted by the pulse generator into a one-cycle pulse that pops the next byte from the FIFO.

4.2 UART receiver

The receiver FSM (Idle, Start, Data, Parity, Stop) works with an edge and bit counter that counts oversampling clock edges within each bit period. The data sampler takes **three samples around the middle of each bit** (at edges $\text{prescale}/2 - 1$, $\text{prescale}/2$ and $\text{prescale}/2 + 1$) and applies a majority vote, which rejects single-sample glitches. Dedicated checkers flag start-bit glitches (the FSM returns to Idle), parity errors and stop-bit errors; `PAR_err` and `STOP_err` are exposed at the top level. A frame is declared valid only when parity and stop checks pass, and back-to-back frames are handled by moving straight from Stop to Start when the line is already low.

4.3 System controller

The controller is a 12-state FSM in the `REF_CLK` domain. In Idle it decodes the opcode of the first frame of a command and branches to the matching sequence. It latches the register address and the ALU function across frames so that a later data byte cannot overwrite them, enables the ALU clock only while an operation is in flight, waits in `ALU_WAIT` until `ALU_Valid`, and pushes the two result bytes into the FIFO, waiting whenever `w_full` is asserted.

State	Role
IDLE	Decode the first frame: 0xAA to <code>WR_ADDR</code> , 0xBB to <code>RD_ADDR</code> , 0xCC to <code>OP_A</code> , 0xDD to <code>ALU_FUNC_NOP</code>
<code>WR_ADDR</code> , <code>WR_DATA</code>	Latch the target address, then write the data byte into the register file
<code>RD_ADDR</code> , <code>RD_WAIT</code>	Issue the read, wait until read data is valid and the FIFO has room, then push the byte into the FIFO
<code>OP_A</code> , <code>OP_B</code>	Write operands into registers 0x0 and 0x1
<code>ALU_FUNC_OP</code> , <code>ALU_FUNC_NOP</code>	Capture the function code with the clock gate enabled (with or without preceding operand writes)
<code>ALU_WAIT</code>	Hold the function code and the clock enable until the ALU reports a valid result
<code>ALU_OUT_L</code> , <code>ALU_OUT_H</code>	Push the low byte, then the high byte of the 16-bit result into the FIFO

4.4 Register file and ALU

The register file has 16 byte-wide registers with a synchronous write port and a registered read port that raises `RD_Valid` with the data. Registers 0x0 to 0x3 are brought out as dedicated outputs: operands A and B feed the ALU, `REG2` carries the UART configuration and `REG3` the TX division ratio. Reset values load the default UART configuration (0x81) and division ratio (0x20).

The ALU consists of four units (arithmetic, logic, comparison, shift) sharing the operands. A small decoder turns `ALU_FUN[3:2]` into per-unit enables, each unit registers its result and a valid flag on `ALU_CLK`, and a top-level multiplexer selects the active unit's 16-bit result. Because the ALU is clocked from the gated clock, it consumes no clock power when idle.

4.5 Clock management

Clock dividers. A counter-based programmable divider with bypass for ratios 0 and 1 (the output then equals the input clock). The dividers are permanently enabled, as specified; one instance feeds RX and one feeds TX. **Clock gating.** A latch-based gate captures the enable while the clock is low and ANDs it with the clock, so the gated clock cannot glitch when the enable changes. The same structure is mapped to an integrated clock-gating cell by synthesis.

4.6 Clock domain crossing strategy

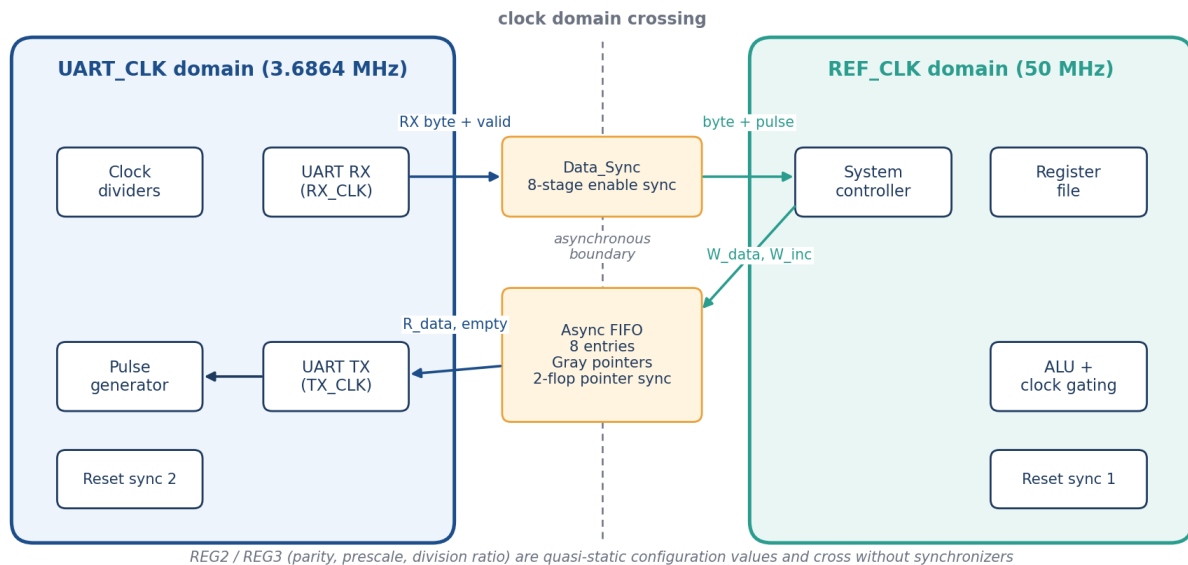


Figure 3. Clock domain crossings between the UART and reference domains.

Crossing	Mechanism
Reset, both domains	Two-stage reset synchronizer: asynchronous assertion, synchronous de-assertion
RX byte to REF_CLK	Data_Sync : multi-stage enable synchronizer (8 stages by default), rising-edge pulse generation, and the data bus is sampled only on that pulse
REF_CLK to TX_CLK (results)	Asynchronous FIFO, 8 entries, Gray-coded pointers, two-flop pointer synchronizers (DF_SYNC); full and empty flags are generated in their own domains
TX busy to FIFO read	Pulse generator in the TX domain converts the busy level into a single read pulse
Config registers to clock dividers	Quasi-static values, recognized by SpyGlass as not needing synchronization

5 Functional verification

Each block has its own testbench in `Testbenches/`, together with a `wave.do` script that sets up a grouped waveform view. The top-level testbench `System_TOP_tb.v` is **self-checking** and acts as the UART master: it drives frames on `RX_IN`, decodes the frames returned on `TX_OUT`, and compares them against expected values. It follows the sequence required by the specification, first configuring registers `0x2` and `0x3` and then issuing register-file and ALU commands. It checks register contents, ALU results, the parity and stop bits of every transmitted frame, that `PAR_err` and `STOP_err` stay low, and that the controller reaches the expected state. It ends with `ALL TESTS PASSED` or `TEST FAILED` and includes a watchdog timer.

5.1 Console results

```

=====
System_TOP testbench
=====

[ Configuration ]
[ 342.1 us] PASS  UART_Config  REG[0x2] = 0x81
[ 681.4 us] PASS  Div_Ratio    REG[0x3] = 0x20

[ Register File write ]
[ 1020.8 us] PASS  RF write     REG[0x5] = 0xa5

[ Register File read ]
[ 1351.2 us] PASS  RF read      REG[0x5] = 0xa5

[ ALU operation with operands ]
[ 1577.2 us] PASS  OP_A stored  REG[0x0] = 0x64
[ 1690.3 us] PASS  OP_B stored  REG[0x1] = 0x0a
[ 2003.0 us] PASS  ALU MUL      A=100 B=10 -> 0x03e8

[ ALU operation without operands ]
[ 2428.7 us] PASS  ALU OR       A=100 B=10 -> 0x006e

[ Frame integrity ]
      PASS  RX flags PAR_err / STOP_err stayed low
      PASS  TX frames: parity and stop bits correct

=====
Checks: 10  Passed: 10  Failed: 0
ALL TESTS PASSED
=====

```

5.2 Waveforms

The waveforms below show the master's frames on RX_IN, the system controller state, the register-file and ALU interfaces, and the responses on TX_OUT.

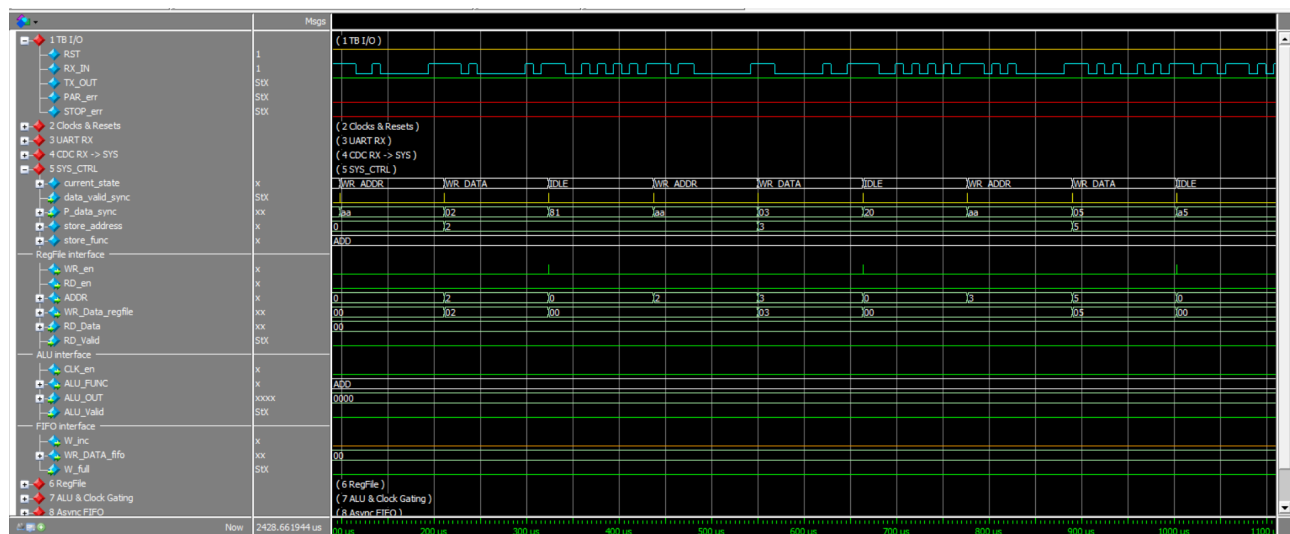


Figure 4. Configuration and register write (90 to 1110 μ s). Three write commands, each 0xAA, address, data: 0xAA 0x02 0x81 (UART configuration), 0xAA 0x03 0x20 (division ratio) and 0xAA 0x05 0xA5 (general-purpose register 5). The controller goes WR_ADDR, WR_DATA, IDLE for each, and WR_en pulses once per command with the latched address.

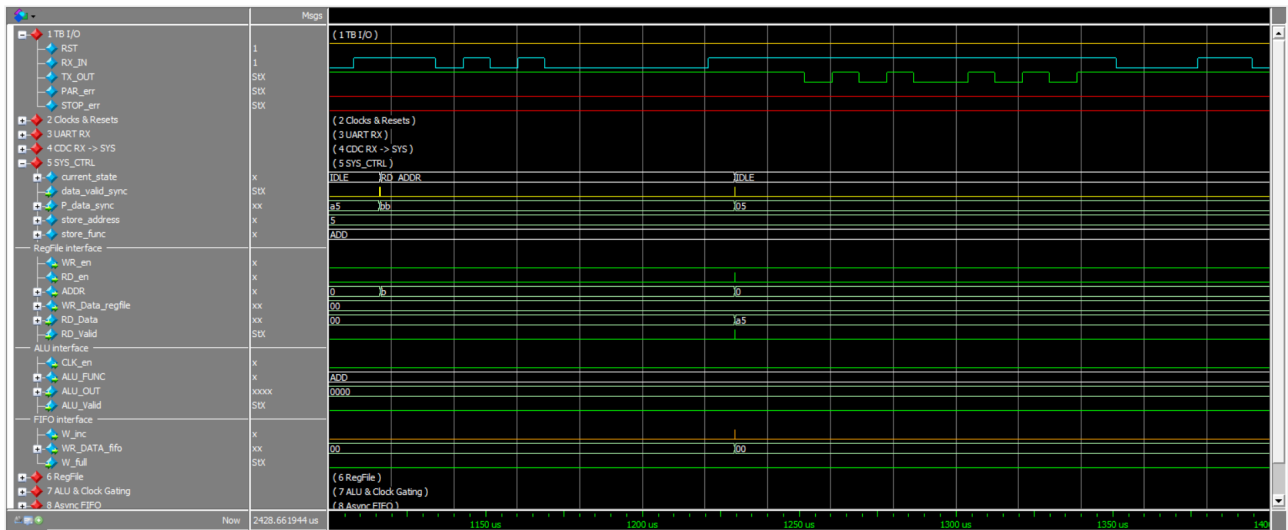


Figure 5. Register read (1100 to 1400 μ s). 0xBB 0x05 reads register 5. RD_Data returns 0xA5, W_inc pushes it into the FIFO, and the response frame starts on TX_OUT.

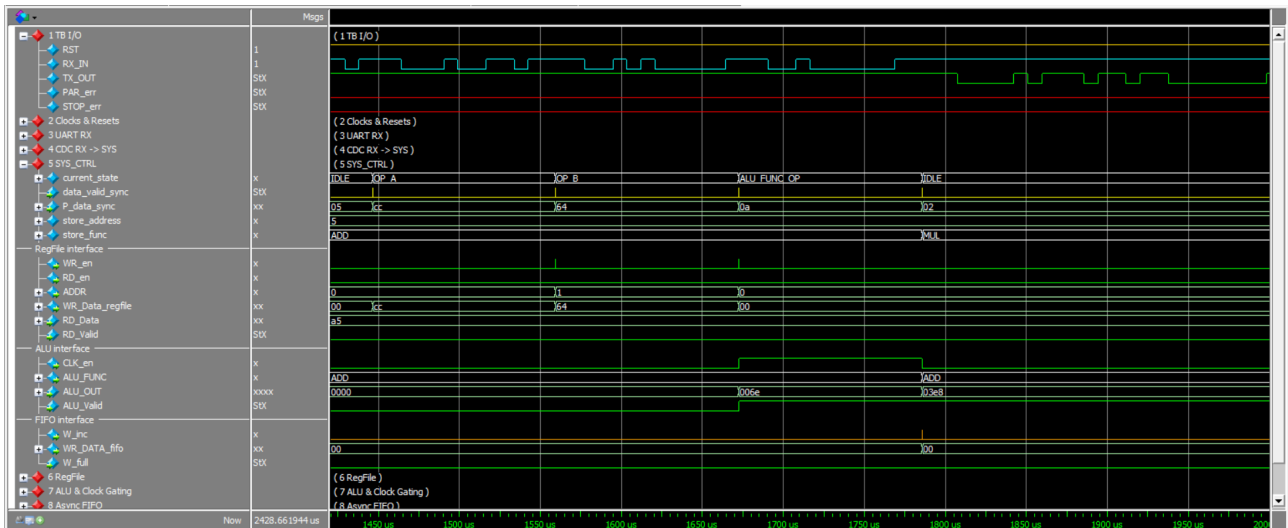


Figure 6. ALU operation with operands (1420 μ s to 2 ms). 0xCC 0x64 0x0A 0x02: operand A (100) goes to REG0, operand B (10) to REG1, then function 0x02 (multiply). CLK_en turns the ALU clock on, ALU_Valid rises, and ALU_OUT settles at 0x03E8 (1000), which is sent back as two bytes.

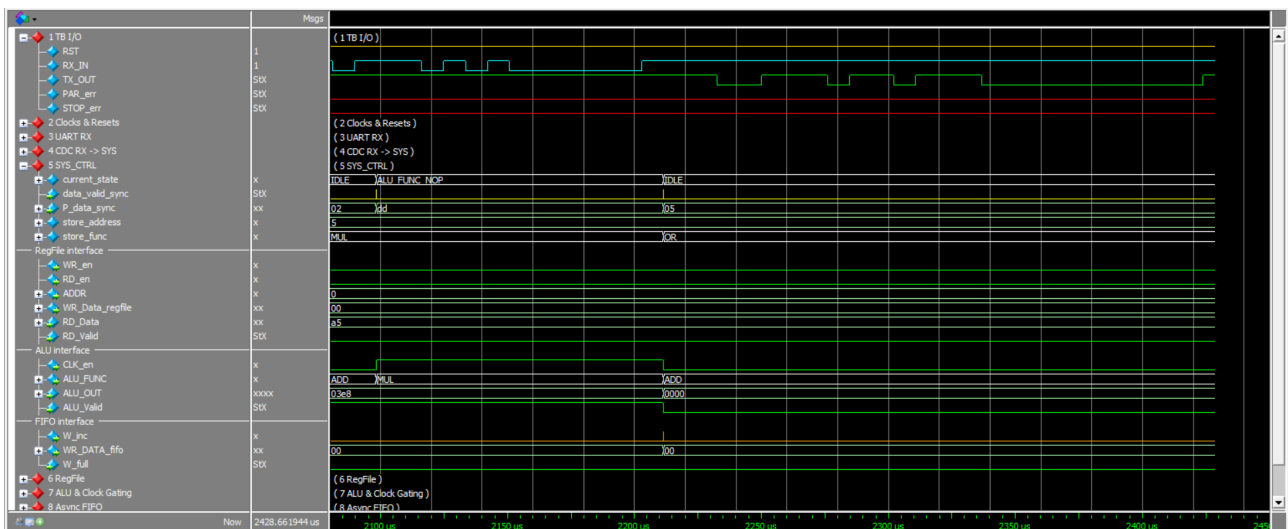


Figure 7. ALU operation without operands (2080 to 2450 μ s). 0xDD 0x05: the controller reuses the operands already in REG0 and REG1 and applies function 0x05 (OR). CLK_en gates the ALU clock on for the operation, and the 2-byte response follows on TX_OUT.

6 Lint and CDC / RDC sign-off

Static checks were run in SpyGlass on the RTL before synthesis. The goals were RTL lint, CDC setup and verification (functional and structural), clock and reset integrity, an abstract view of the design, and reset domain crossing (RDC) analysis. Violations that are by design were waived with documented waivers.

Check	Result
Lint (RTL)	3 messages, 1 waived; the others are informational
CDC verify	52 messages, 6 waived; no Error or Warning severity
CDC verify (structural)	46 messages, 3 waived
Unsynchronized scalar / vector crossings	0 / 0
Synchronized scalar / vector crossings	7 / 15
Glitches on synchronized control paths	0
Data loss on fast-to-slow control crossings	0
Resets with reset synchronizers / without	2 / 0
Reset domain crossings (RDC)	0

The result confirms the crossing strategy of section 4.6: every crossing from one domain to the other goes through one of the intended synchronizing structures, and no signal crosses unsynchronized. The configuration registers that feed the clock dividers are quasi-static and are recognized as such.

7 Logic synthesis

The design was synthesized with Design Compiler (O-2018.06-SP1) against the TSMC 0.13 μm library at the slow corner for setup and the fast corner for hold, with the following constraints.

Constraint	Value
REF_CLK	20 ns period (50 MHz)
UART_CLK	271.27 ns period (3.6864 MHz)
Generated clocks	ALU_CLK (gated REF_CLK), RX_CLK (/1), TX_CLK (/32)
Clock uncertainty	0.2 ns setup, 0.1 ns hold
I/O delays	20% of the clock period on inputs and outputs
Clock groups	REF_CLK / ALU_CLK and UART_CLK / RX_CLK / TX_CLK declared asynchronous

7.1 Area and power

Metric	Value
Total cell area	25,680 μm^2
Combinational / non-combinational area	16,290 / 9,390 μm^2
Cells	2,282 (1,857 combinational, 410 buffers and inverters)
Total power (SS, 1.08 V, 125 °C)	0.363 mW (switching 0.111, internal 0.234, leakage 0.017)
Constraint violations	none

7.2 Timing

Clock group	Setup slack	Hold slack
ALU_CLK (50 MHz)	0.05 ns	1.38 ns
REF_CLK (50 MHz)	6.96 ns	0.52 ns
UART_CLK / RX_CLK (3.6864 MHz)	265.97 / 261.39 ns	0.52 / 1.19 ns
TX_CLK (115.2 kHz)	8671.88 ns	0.61 ns

All paths meet timing. The critical path runs from the register file to the ALU on the gated clock, which is why ALU_CLK has by far the smallest setup slack after synthesis. Power figures use propagated switching activity, not simulation-annotated activity.

8 Design for test

Scan insertion converts every flip-flop in the design into a multiplexed-flip-flop scan cell and stitches the cells into four scan chains. A scan-ready top level (SYS_TOP.v) adds the test-mode multiplexers: in test mode the four functional clocks (REF_CLK, UART_CLK, RX_CLK, TX_CLK) and the three reset paths are driven from a single scan_clk and scan_rst, so all scan cells can be shifted with one clock and the design starts from a controlled state.

Metric	Value
Scan style	Full scan, multiplexed flip-flop, no clock mixing
Scan chains	4 (94 / 94 / 93 / 93 cells), 374 scan cells
Scan ports	SI[3:0], S0[3:0], SE, test_mode, scan_clk, scan_rst
Estimated test coverage	99.53%
Cells / area after scan	2,710 cells, 31,125 μm^2 (+21.2% over synthesis)
Power after scan	0.516 mW (SS corner)
Setup slack	18.44 ns (ALU_CLK), 68.79 ns (SCAN_CLK)
Hold slack	1.01 ns (ALU_CLK), 0.79 ns (SCAN_CLK)

The area increase of 21.2% is the usual cost of multiplexed scan flip-flops and the test-mode multiplexers. The post-DFT schematic from Design Vision below shows the top-level blocks together with the scan clock and scan reset multiplexers.

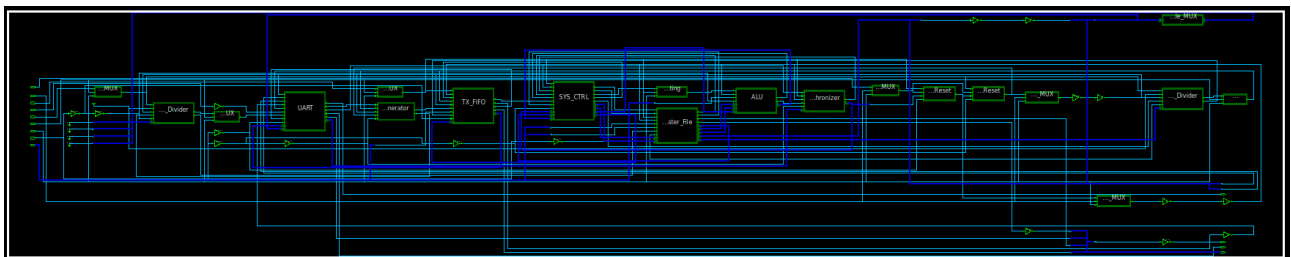


Figure 8. Post-DFT schematic of SYS_TOP (Design Vision).

9 Place and route

The scan netlist is placed and routed in Cadence SoC Encounter 8.1 using three constraint modes (functional, scan and capture). Setup is checked at the slow corner (SS, 1.08 V, 125 °C) and hold at the fast corner (FF, 1.32 V, -40 °C), which gives six analysis views.

9.1 Flow steps

Step	Script	Setup
Import	des_import.tcl, import/MMMC.tcl	TSMC 0.13 μm LEF and capacitance tables, 3 libraries, 6 analysis views
Floorplan	floorplan.tcl	240.47 $\times$ 160.47 μm die, 6 μm core-to-die margin
Placement	placement.tcl	placeDesign with in-place and pre-place optimization, tie cells, power-net connection
Clock tree synthesis	cts.tcl	clockDesign from Clock.ctstch, reports in clock_report/
Routing	routing.tcl	NanoRoute global and detail routing with via and wire optimization, up to metal 6
Finishing	chip_finish.tcl	Filler insertion (FILL1M to FILL64M, 552 filler cells)
Outputs	outputs_gen.tcl	GDSII, netlist (with and without power pins), SDF and SPF in pnr/export/

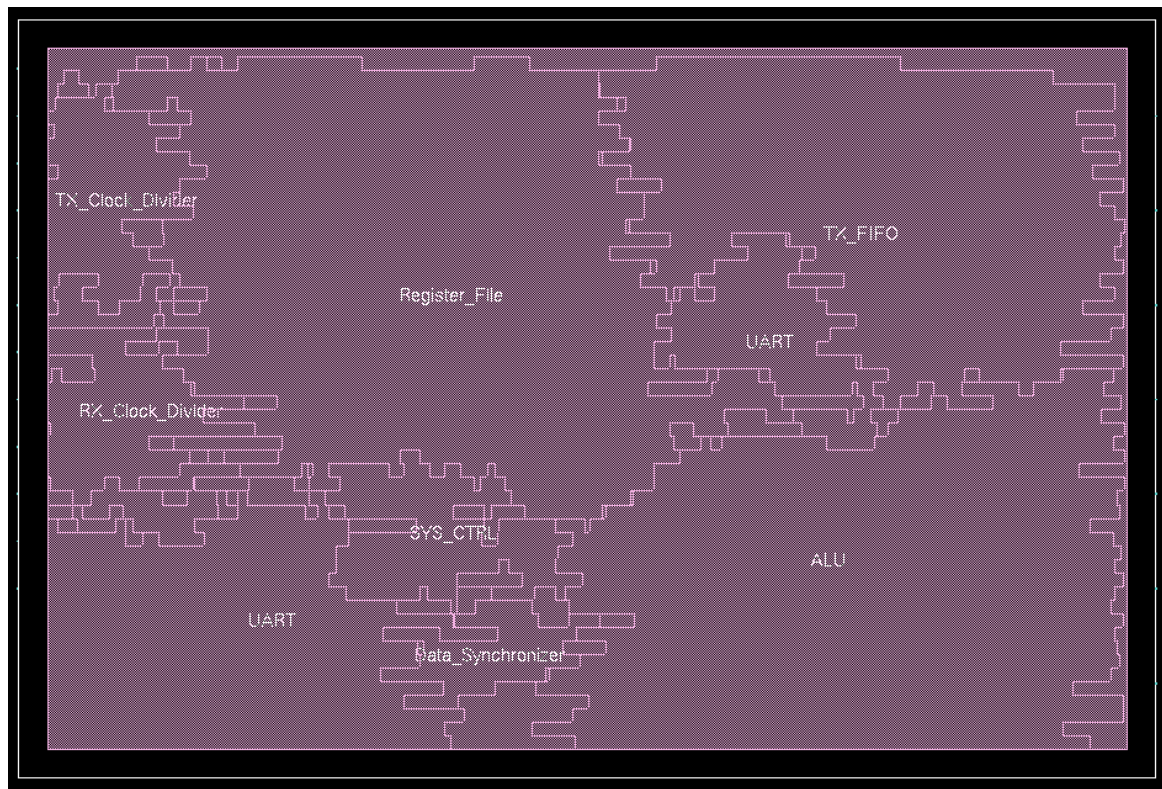


Figure 9. Placed design colored by module: clock dividers, register file, TX FIFO, UART, system controller, ALU and data synchronizer.

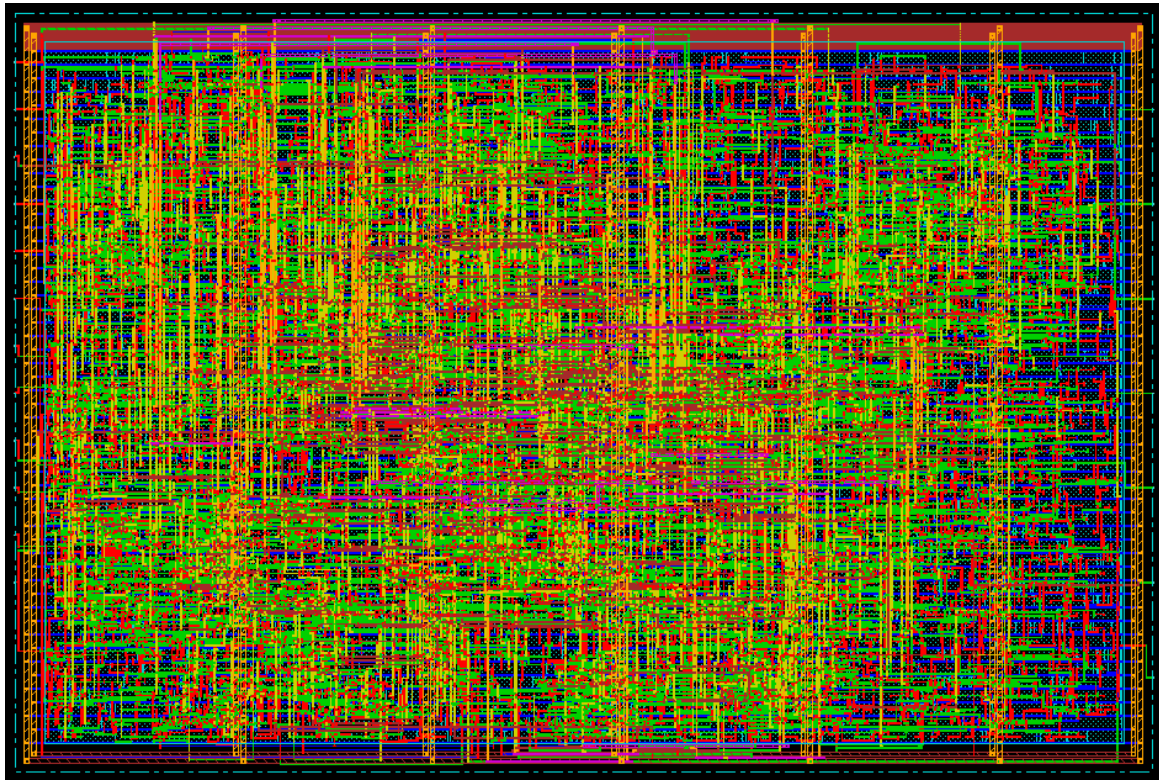


Figure 10. Final routed layout in SoC Encounter.

9.2 Post-route results

Metric	Value
Setup WNS / TNS	+3.431 ns / 0.000 ns, 0 violating paths out of 1,062
Hold WNS / TNS	+0.019 ns / 0.000 ns, 0 violating paths
Design rule violations (max cap / max transition / max fanout)	0 / 0 / 0
Geometry (DRC)	No violations found
Connectivity	No problems or warnings
Process antenna	No violations found
Cell density	93.4% before filler insertion, 98.53% after
Total power (SS, 1.08 V)	0.856 mW (internal 0.566, switching 0.268, leakage 0.022)

Path group	Setup WNS	Hold WNS
Register to register	3.431 ns	0.019 ns
Input to register	15.412 ns	0.113 ns
Register to output	77.152 ns	4.366 ns
Clock gating	17.639 ns	0.520 ns

9.3 Clock tree

Clock tree synthesis produced 14 subtrees driving 372 sinks through 76 clock buffers over 18 levels. Skew is 229 ps in the setup views and 89.4 ps in the hold views, against a 200 ps target, so the setup views are slightly over the skew target; timing nevertheless closes after routing, with positive slack in every path group. Power is computed with a 0.2 primary-input activity and no annotated switching activity.

10 Formal equivalence checking

Formality compares each netlist against the golden RTL after every transformation, which guarantees that synthesis, scan insertion and place and route did not change the functionality. For the DFT and place-and-route comparisons the reference is the RTL together with the scan-ready top level.

Comparison	Compare points	Passing	Failing	Aborted / unverified	Result
RTL vs. post-synthesis netlist	369	369	0	0	Verification SUCCEEDED
RTL with DFT muxes vs. post-DFT netlist	378	378	0	0	Verification SUCCEEDED
RTL with DFT muxes vs. post-route netlist	378	378	0	0	Verification SUCCEEDED

Compare points after synthesis are 3 ports, 365 flip-flops and 1 latch (the clock-gating latch); after DFT and after place and route they are 3 ports, 374 flip-flops and 1 latch. For the last two comparisons `test_mode` and `SE` are held at 0 and the scan data ports `SI` and `S0` are excluded, since they do not exist in the functional RTL.

11 Cross-stage summary and limitations

11.1 Summary across the flow

Stage	Area / cells	Total power	Timing	Checks
Synthesis	25,680 μm^2 , 2,282 cells	0.363 mW	All paths met	No constraint violations; Formality 369 / 369
DFT	31,125 μm^2 , 2,710 cells	0.516 mW	All paths met	99.53% test coverage; Formality 378 / 378
Place and route	240.47 $\times$ 160.47 μm die, 93.4% density before filler	0.856 mW	Setup WNS +3.431 ns, hold WNS +0.019 ns	DRC, connectivity and antenna clean; Formality 378 / 378

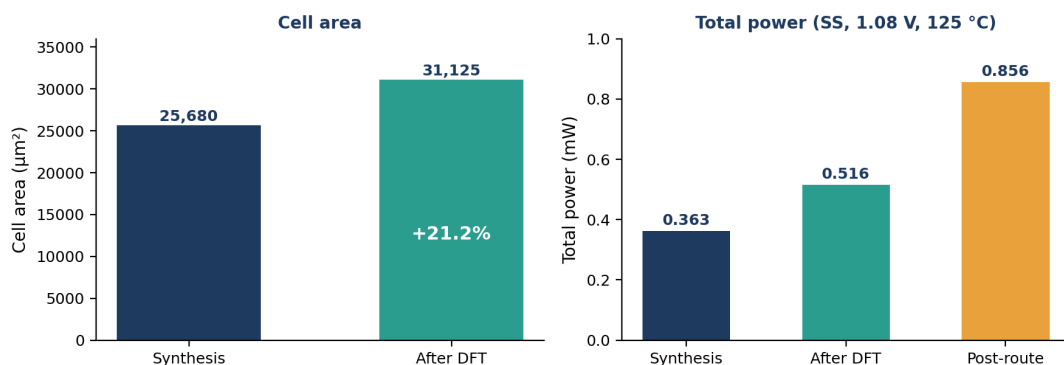


Figure 11. Cell area (synthesis and after DFT) and total power through the flow. The post-route power also includes the clock tree and routing parasitics.

11.2 Limitations and observations

- **Clock tree skew.** Post-CTS skew is 229 ps in the setup views against a 200 ps target. Timing still closes after routing with positive slack in all path groups.
- **Tight ALU timing at synthesis.** The register-file-to-ALU path on the gated clock had only 0.05 ns of setup slack after synthesis. It was met, and post-route reg-to-reg setup WNS is +3.431 ns.
- **Power estimation.** Power numbers use default or propagated switching activity, not activity annotated from simulation, so they are estimates rather than sign-off values.
- **Test coverage.** Estimated test coverage is 99.53%, not 100%.
- **Post-layout simulation.** The design is verified by RTL simulation and formal equivalence at each stage; gate-level simulation of the routed netlist with the exported SDF is not part of this report.

11.3 Possible extensions

- Gate-level simulation of the post-route netlist with back-annotated SDF to confirm timing at the slow and fast corners.
- Power analysis with switching activity from the system testbench (VCD or SAIF) for sign-off-quality numbers.
- Constrained-random UART stimulus with injected parity and stop errors, and functional coverage of every ALU function.
- Tightening the clock tree to meet the 200 ps skew target in the setup views.

12 Conclusion

The UART-based master-slave system controller was designed, verified and implemented through the full RTL-to-GDSII flow. The two clock domains are connected only through verified synchronizing structures, with no unsynchronized crossings in the CDC and RDC analysis. Simulation passes all 10 self-checking tests, synthesis and place and route meet timing with no violations, the scan insertion reaches 99.53% estimated coverage, and formal equivalence succeeds after synthesis, after DFT and after place and route. The routed layout is clean in geometry, connectivity and antenna checks.

Appendix A Repository structure

— RTL/	Source, one folder per block
— System Top/	System_TOP.v
— System Controller/	System_Controller.v
— Register File/	register_file.v
— Signed ALU/	alu_top, arithmetic / logic / cmp / shift units, decoder
— UART_TOP/	UART_TOP.v (TX + RX wrapper)
— UART_TX/	FSM, MUX, Serializer, Parity_Calculator, UART_TX_top
— UART_RX/	FSM_RX, Edge_Bit_Counter, Data_Sampler, Deserializer, Start / Parity / Stop checkers, UART_RX_top
— Asynchronous FIFO/	ASYC_FIFO, FIFO_WR, FIFO_RD, FIFO_MEM_CONTROL, DF_SYNC
— Clock Divider/	Clock_Divider.v
— Clock Gating/	Clock_Gating.v
— DFT MUX/	mux2X1.v (scan clock / scan reset mux)
— Prescale MUX/	Prescale_MUX.v
— Pulse Generator/	Pulse_Generator.v
— Data Synchronizer/	Data_Sync.v
— Reset Synchronizer/	Reset_SYNC.v
— Testbenches/	Per-block and top-level testbenches, wave.do, run.do.txt
— Linting & CDC/	
— LINT/	RTL lint reports
— CDC/	CDC, RDC, setup, clock/reset integrity, abstract views
— Synthesis/syn/	Design Compiler scripts, constraints, netlist, SDC/SDF, reports
— Design For Test (DFT)/	Scan insertion script, scan-ready SYS_TOP.v, netlist, reports
— Place & Route (PnR)/	
— DFT/	Scan netlist and SDC/SDF used as the PnR input
— pnr/	Encounter scripts, timing / clock / power reports, GDS, netlist, SDF
— std_cells/	Libraries, LEF and capacitance tables
— Formal Verification/	
— Formality Post Synthesis/	RTL vs. synthesized netlist
— Formality Post DFT/	RTL with DFT muxes vs. scan netlist
— Formality Post PnR/	RTL with DFT muxes vs. routed netlist
— docs/	Block diagram, waveforms, schematic, PnR screenshots

Appendix B Tools and technology

Purpose	Tool or library
Simulation	ModelSim / Questa
Lint, CDC, RDC	SpyGlass
Logic synthesis and scan insertion	Synopsys Design Compiler O-2018.06-SP1, Design Vision for schematics
Formal equivalence	Synopsys Formality
Place and route	Cadence SoC Encounter 8.1
Standard cells	TSMC 0.13 μm scmetro_tsmc_cl013g, regular Vt; slow (1.08 V, 125 °C), typical and fast (1.32 V, -40 °C) corners